

Course Description: Data Structures and Database Systems

This course integrates the fundamental principles of data structures and database systems, two core pillars of computer science, equipping students with the theoretical knowledge and practical skills to design, implement, and manage efficient data storage and retrieval solutions. The course balances theoretical foundations with hands-on practice, enabling students to understand how data is organized, processed, and persisted in modern computing environments.

In the data structures segment, students will explore essential data organization methods, including arrays, linked lists, stacks, queues, trees (binary trees, AVL trees, B-trees), graphs, hash tables, and heaps. They will learn to analyze the time and space complexity of different data structures, master the selection and application of appropriate structures for specific problem scenarios, and implement basic data structure operations through programming exercises. Emphasis is placed on understanding the trade-offs between efficiency, usability, and scalability in data structure design.

The database systems segment focuses on the design, implementation, and management of relational databases, the most widely used database model in industry. Students will study database fundamentals, including entity-relationship (ER) modeling, relational algebra, structured query language (SQL) for data manipulation (insert, update, delete) and querying, database normalization to eliminate data redundancy, and transaction management (ACID properties). Additionally, the course introduces key database concepts such as indexing, query optimization, and basic database administration, providing insights into how databases support real-world applications.

Throughout the course, students will engage in practical assignments, programming projects, and database design exercises. They will apply data structures to solve complex algorithmic problems and use SQL to design and interact with relational databases, gaining hands-on experience in building data-driven applications. The course also touches on the connections between data structures and databases—how efficient data structures underpin database performance, and how database systems leverage data structures to optimize storage and retrieval.

By the end of the course, students will be able to: (1) Analyze and select appropriate data structures for diverse computational tasks; (2) Implement and manipulate core data structures using a programming language; (3) Design relational databases using ER models and normalization techniques; (4) Write efficient SQL queries to retrieve, modify, and manage data;

(5) Understand the principles of database transaction management and query optimization; (6) Apply theoretical knowledge to solve real-world data management problems.

Prerequisites: Basic programming skills (e.g., in Python, Java, or C++), fundamental understanding of computer organization and algorithms. This course is suitable for undergraduate students majoring in Computer Science, Software Engineering, Information Systems, and related fields, laying a solid foundation for advanced courses in data engineering, machine learning, and distributed systems.

(注：文档部分内容可能由 AI 生成)